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# MINDSET 1.0: Methodology and Use*

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**Abstract**

This paper presents MINDSET 1.0, Model of Innovation in Dynamic Low-Carbon Structural Economic and Employment Transformations, a multi-regional input-output model covering 163 regions and 120 sectors. Introducing endogenous adjustments of production, consumption, and trade, the model is designed to assess the short-term macroeconomic, sectoral, and labor market impacts of climate change, and of climate, and development and trade policies. By connecting to labor force and household survey data, MINDSET disaggregates employment and consumption impacts across income strata, skill levels, occupation types, and sub-national regions - providing the distributional granularity needed for active labor market policy design and targeted government support in the context of resilient and low-carbon structural transformation.

**JEL Classification:** C51, C67, E12, J23, Q43, Q54, Q56, H23, O44

**Keywords:** macroeconometric input-output model, climate policy, climate change impacts, labor demand, distributional effects

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The MINDSET 1.0 model code is open-source and under continued development, this document reflects the state of the model as of December 2025. The latest version of the model's code is available at `https://github.com/cpmodel/MINDSET_module/`.

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# 1 Introduction

MINDSET is a global simulation model used to evaluate the impact of climate change, climate policies, and other development and trade policies on macroeconomic, sectoral, and labor market outcomes. MINDSET stands for Model of Innovation in Dynamic Low-Carbon Structural Economic and Employment Transformations. This documentation presents MINDSET, a multi-regional input-output model. By introducing endogenous adjustments of production, consumption, and trade, the model predicts policy-induced shifts across all sectors and regions. It is parametrized with sector- and country-specific elasticities, including energy efficiency and fuel switching, household income and demand, trade, output-driven investment (and investment converters), and output-employment. This relatively basic framework has been succeeded by the development of MINDSET 2.0, which solves annually based on its own parametrization framework.

The goal of MINDSET is to provide actionable policy advice tailored to country-specific opportunities and risks, while adopting a whole-of-economy and people-centered strategy. To effectively advise on the design of sustainable and resilient development policies, it is essential to quantify the sectoral implications of policy-induced structural changes. While development and climate policies can bring positive effects for employment at an economy-wide level, impacts can vary substantially across sectors, income groups, and skill levels. Understanding the short- to medium-term labor market impacts and their distribution across winners and losers is paramount for achieving positive development outcomes and gaining public support.

The core motivation for developing MINDSET is to provide a tractable, user-friendly, and cost-efficient framework for assessing the employment effects of climate and development strategies, and their distributional impacts on workers, sectors, and regions. Model simulations can assess a broad range of fiscal, price-based, and regulatory policies, as well as climate change impacts (Table 1). The model has a high level of disaggregation, covering 163 regions, with 120 sectors in each region. Hence, it supports detailed and comprehensive sector- and country-specific, single- or multi-country policy and climate scenarios. Linked to sectoral models, macroeconomic and distributional impacts of within-sector transitions can be analyzed.[1] MINDSET's outputs include standard macroeconomic indicators, sectoral employment, greenhouse gas (GHG) emission levels, and prices (Table 1). By linking to countries' labor force and household survey data, impacts on workers and consumers can be analyzed across income strata, skill levels, occupation types, and sub-national provinces, providing relevant information for active labor market and re-skilling interventions and sectoral or regional government support funds.

MINDSET diagnostics have supported jobs-related and distributional analyses in over 30 country engagements with a focus on the five areas prioritized for job creation by the World Bank - infrastructure and energy, agribusiness, healthcare, tourism, and local manufacturing. For example, MINDSET diagnostics have been used to assess the sectoral job and equity effects of repurposing LPG subsidies towards healthcare and insurance in Morocco (PforR); how renewable energy investments shift job creation across occupations and income groups in Moldova; how climate shocks to tourism affect women's jobs in local services in Namibia; how expanding copper exports could generate more jobs for Mongolia, than will be lost due to shrinking coal export markets; how investments in resilient infrastructure would boost jobs in the construction sector and along local supply chains in the Philippines; how climate shocks to crop yields would impact rural livelihoods and undermine poverty reduction efforts in Laos; or how the European Union's Carbon Border Adjustment Mechanism could threaten local manufacturing and supplying services sector jobs in North Macedonia (see Appendix 6.1 for a full list of published country engagements).

MINDSET adds value as a practical and versatile tool for applied policy analysis. Its design combines flexibility with analytical depth, making it suitable for supporting decision-making, including in lower-income countries. MINDSET sits between a pure input-output model and a full macroeconometric model which entails specific strengths and weaknesses:

- **Capturing short-term adjustment frictions:** MINDSET combines multi-regional input-output analysis with demand-led macroeconometric modeling, capturing short-term country-specific frictions. The

[1] For example, MINDSET has been used to analyze the wide range of distributional and sectoral impacts of low-carbon measures in agriculture, transportation, or housing based on technology-specific inputs from the energy-sector model TIMES, or the World Bank's Electricity Planning Model (EPM).

| Policy and climate impact scenarios | Metrics of interest for policy advice |
|---|---|
| • Acute climate impacts: e.g., floods, storms, or wildfires, quantified by direct production losses<br>• Chronic climate impacts: e.g., heat stress or loss of productivity<br>• Carbon prices: commodity or carbon tax, subsidy, and tariff changes, differentiated by commodity & sector<br>• Carbon Border Adjustment Mechanism (CBAM): single- or multi-country trade scenarios, differentiated by sectors<br>• Income support or tax reforms<br>• Sectoral public spending, cross-funded or additional<br>• Sectoral public investment, cross-funded or additional<br>• Electricity and transport sector policies (with sectoral models) | • Sectoral job losses/gains by:<br>– Occupations & skill level<br>– Income groups<br>– Gender & age<br>– Spatial: province/region<br>• GDP, output (sectoral: GVA, output)<br>• Tax revenue collection (by sector)<br>• Emissions effects (GHG)<br>• Sectoral consumer price changes & consumption incidence across income groups<br>• Competitiveness and sectoral import and export changes |

Table 1: Policy and climate impact scenarios and results metrics

model simulates the direct economic and distributional effects of granular policies or climate shocks *before economies have fully reallocated resources*, making it particularly suited for contexts where institutional rigidities, incomplete information, or imperfect markets prevent fully rational, intertemporal optimization. MINDSET assumes spare economic capacity and involuntary unemployment, and is thus able to represent scenarios where derived demand is met under slack conditions. There are no 'crowding out' effects in which climate- or development-related investment automatically leads to lower investment elsewhere in the economy. However, in cases where capacity constraints may bind, such as immobility of labor or climate damages to capital, additional off-model assumptions are required in the scenario assumptions to represent these supply-side constraints and their cascading impacts along downstream supply chains (see subsubsection 5.2.2).

- **Tracing intersectoral and cross-country linkages and spillovers:** Built on a detailed network of interindustry and trade linkages linked to countries' energy balances, the model traces *sectoral and cross-border spillovers, third-country effects, and trade policies*, capturing how shocks and policy changes propagate through global value chains and affect competitiveness across interconnected economies. Results capture direct, indirect, and induced supply-chain impacts, which together constitute fiscal multipliers, with a particular focus on jobs.
- **Substantial detail in labor market and consumer outcomes:** Distributional effects are represented through disaggregation by occupation, income level, location, and gender, when linked to household survey data (section 4).[2]
- **Providing transparency and user-friendliness:** MINDSET's simplicity makes the model accessible and its results extremely transparent and understandable for policymakers. By solving iteratively rather than simultaneously, MINDSET preserves the stepwise transmission of shocks through the economy,

[2]As discussed in section 4, the distributional routines in MINDSET are simplified and the sectoral and labor market detail is intended to facilitate linkages with more sophisticated tools such as the World Bank's Microsim model (Bourguignon, Bussolo, and Pereira da Silva, 2008).

enabling users to *trace results back to their drivers*. This bottom-up shock propagation highlights the role of specific sectors and channels in driving overall outcomes. The model is built entirely on *open-source data* structured in accessible spreadsheets and implemented in Python, promoting country-specific adjustment of assumptions and capacity building without reliance on proprietary tools.

- **Ensuring broad applicability and cost-effectiveness:** MINDSET's *relatively simple setup* provides two practical advantages: first, it is broadly applicable to low-income and data-scarce contexts, *bringing sectoral and distributional analysis to a wider range of countries*; and second, it allows for *rapid, low-cost applications*, as country-specific baseline setup and data adjustments are not time consuming and the broad range of supported scenarios as well as user-friendly parameter input sheets facilitate speedy sensitivity analyses.

The remainder of this documentation introduces the model's input-output framework and data (section 2), the included model mechanisms (section 3), and the distributional modules (section 4). A discussion section (section 5) provides a comparison of MINDSET results to those of other macromodels and discusses some of its limitations.

# 2 The multiregional input-output setup and data

The core of the model is a multiregional input-output (MRIO) system, combined with the accounting identities defined in the System of National Accounts (SNA) framework. This globally balanced system of monetary flows captures all inter-country inter-industry flows of the goods and services used as intermediate inputs to production and for final demand from households, government, and businesses. Monetary flows are augmented with physical flows of energy, or fuels, and emissions. This section introduces basic MRIO setup, the relevant accounting identities, and describes the construction of the corresponding physical flows.

## 2.1 Monetary flows in the MRIO system

Multi-regional input-output tables provide a globally balanced view of the domestic and international flows of goods and services, based on national-level data of all countries. The MRIO tracks bilateral inter-industry flows which are used as intermediate inputs into production and those to meet final demand. Recently developed MRIO systems[3] are constructed to represent countries' officially published statistics to the maximum extent possible: the main economic aggregates, such as GDP, final demand, and exports and imports, are benchmarked to countries' SNA, particularly the expenditure, output, and income accounts (cf. Lequiller and Blades, 2014). Production recipes, output, and bilateral trade by industry are based on countries' input-output tables and trade statistics (Yamano et al., 2023).

The main elements of MRIO tables comprise a matrix of domestic and cross-border sector-to-sector monetary flows of intermediate inputs $\mathbf{Z}$), a matrix capturing output used to meet total final demand ($\mathbf{y}$), a vector of total output by country and industry ($\mathbf{x}$), and vectors representing value added generated in production by country and industry ($VA$). Final demand is differentiated between households ($C$), government ($G$), and fixed capital formation ($FCF$), or 'investment'. Value-added components are typically compensation of employees (wages), taxes less subsidies on production, and gross operating surplus (profits). Taxes minus subsidies on both intermediate and final products are reported separately to ensure that intermediate and final demand are recorded at basic prices. The row and column sums each represent total sectoral output (Miller and Blair, 2009; Yamano et al., 2023).

MRIO models extend this concept to achieve a representation of relationships between economic sectors in several countries at the same time (Miller and Blair, 2009). The matrix of intermediate inputs ($\mathbf{Z}$) is four-dimensional as it specifies country-country-industry-industry flows which thus incorporate imports and exports in production. The final demand matrix, or vectors, are three-dimensional, differentiating the

[3]EORA (Lenzen et al., 2013; Lenzen et al., 2012), EXIOBASE (Stadler et al., 2018), GTAP (Aguiar et al., 2022) and WIOD (Timmer et al., 2015), Eurostat (Eurostat, 2022), OECD (OECD, 2022), Asian Development Bank (Asian Development Bank, 2022), ECLAC (Economic Commission for Latin America and the Caribbean (ECLAC), 2020).

country of origin and of consumption for each sector (or product). The vectors of value-added items are equally three-dimensional.

### 2.1.1 Main input-output matrix calculations

For a single country, the sectoral output vector $\mathbf{x}$ of industries $n$, the intermediate inputs matrix $\mathbf{Z}$, and the aggregate final demand vector $\mathbf{y}$ can be expressed as:

$$\mathbf{x} = \mathbf{Z}\mathbf{i} + \mathbf{y} \tag{1}$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{Z} = \begin{bmatrix} z_{11} & \cdots & z_{1n} \\ \vdots & \ddots & \vdots \\ z_{m1} & \cdots & z_{mn} \end{bmatrix} \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \tag{2}$$

where industries $1, \ldots, m$ are producing sectors and sectors $1, \ldots, n$ are supplying industries, while $n = m$, such that the matrix is symmetric, and the elements $z_{mn}$ denote the monetary value of intermediate inputs from sector $n$ used by sector $m$. $\mathbf{i}$ denotes a column vector of ones such that multiplying the matrix by $\mathbf{i}$ yields a column vector of row sums of that matrix. Total output of each sector $n$ equals the row-sum of supplied intermediate and final demand, both from domestic production and for exports (Miller and Blair, 2009).

The technology matrix $\mathbf{A}$ is obtained by dividing each column of the intermediate flows matrix $\mathbf{Z}$ by the corresponding sector's total output $x_n$, written as

$$\mathbf{A} = \mathbf{Z}\,\hat{\mathbf{x}}^{-1} \tag{3}$$

where $\hat{\mathbf{x}}$ denotes the diagonal matrix with sectoral outputs $x_n$ on its diagonal:

$$\hat{\mathbf{x}} = \begin{bmatrix} x_1 & 0 & \cdots & 0 \\ 0 & x_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & x_n \end{bmatrix} \tag{4}$$

The inverse $\hat{\mathbf{x}}^{-1}$ has $1/x_n$ on its diagonal, ensuring that each column of $\mathbf{Z}$ is normalized by the output of the corresponding sector.

Hence, Equation 1 can be rewritten as:

$$\mathbf{x} = \mathbf{A}\mathbf{x} + \mathbf{y} \tag{5}$$

such that elements $a_{mn}$ of matrix $\mathbf{A}$ denote technical, or direct-input, coefficients, i.e., how much of sector $n$'s output is used directly *per unit of output* of sector $m$.

The total input requirements from sector $n$, throughout the whole upstream value chain, to produce one unit of final demand in sector $m$ are expressed in the Leontief matrix, or Leontief inverse, $(\mathbf{I} - \mathbf{A})^{-1}$, where $\mathbf{I}$ is an identity matrix (cf. Leontief, 1986; Miller and Blair, 2009):

$$\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{y} \tag{6}$$

This **Leontief quantity model** links final demand to output by industry via the identity matrix and the input coefficient matrix. The Leontief quantity model delivers a *demand-pull effect*, or the output (or

demand) multiplier, i.e., the production required from each sector and country to satisfy a given final demand. In MINDSET, the Leontief quantity model is used to simulate the economic impacts of demand shocks, such as increased household consumption.

Complementing this demand-pull perspective, the **Leontief price model** links costs of production to output prices. It relates value-added shares ($\mathbf{v}$), comprising compensation of employees, net operating surplus, and taxes on production, as well as product tax shares ($\mathbf{t}$), to prices ($\mathbf{p}$):

$$\mathbf{p} = (\mathbf{v} + \mathbf{t}) \times (\mathbf{I} - \mathbf{A})^{-1} \tag{7}$$

In the base year, since variables are expressed in nominal terms, the price vector $\mathbf{p}$ is a vector of ones.

When production costs (e.g., wages, taxes, or subsidies) change exogenously under cost-push assumptions, the resulting effect on the overall price level can be calculated by pre-multiplying the Leontief inverse[4].

The reverse logic can be expressed in the **supply-side Ghosh quantity model**. It relates each unit of sectoral output entering value chains upstream to producing sectors downstream, and finally to a unit of final demand. The method is implemented by shifting from a column-oriented (vertical) representation of the model to a row-oriented (horizontal) one. Instead of dividing each column of the transaction matrix $\mathbf{Z}$ by the total output of the corresponding purchasing sector, each row is divided by the total output of the corresponding supplying sector. This produces the Ghoshian inverse matrix $(\mathbf{I} - \mathbf{B})^{-1}$, which contains the direct output coefficients used in the Ghosh model (Miller and Blair, 2009, p. 544 ff). In MINDSET, the Ghosh quantity model is used to simulate the downstream trickle-down impacts of supply shocks, or supply constraints, such as from physical damages from extreme weather events (Hallegatte, 2014; Pollitt and Petrauskaite, 2024) or other disruptions to production (e.g., Rio-Chanona et al., 2020 on COVID supply constraints). In these scenarios, final demand is forced down proportionately to the supply shock, reversing the logic of the demand-driven MRIO setup.

MINDSET uses these structural identities of the national accounting framework and calculations of the input-output system as its foundation (Lequiller and Blades, 2014; Lewney, Pollitt, and Mercure, 2019). Hence, knowing final demand allows for the calculation of total output. This assumes that the accounting identities are valid and that intermediate, or 'technical', coefficients remain constant[5]. Gross value added ($GVA$) and its components can then be directly computed from this setup if the $GVA$ ratios remain constant.

### 2.1.2 Data

In its default application, MINDSET uses MRIO data from the open-source Global Resource Input-Output Assessment (GLORIA) database (Lenzen et al., 2017; Lenzen et al., 2021). GLORIA has 163 regions, 120 sectors each, supply-use transactions, final demand, value added in five valuations (basic prices, trade margins, transport margins, taxes on products, subsidies on products). Final demand is differentiated between five main agents and categories: households, government and gross fixed capital formation (FCF), as well as non-profit institutions, and changes in inventories. Value-added categories are compensation of employees (i.e., salaries), taxes and subsidies on production, net operating surplus, net mixed income, and the consumption of fixed capital (i.e., depreciation of assets). This setup provides an excellent data basis, although model users are encouraged to cross-check with or integrate data for specific countries of interest into this setup.

## 2.2 Physical energy and emissions flows

Similar to the matrices of monetary transactions, MINDSET tracks physical flows of energy and energy-borne emissions, in particular for analyzing the economic impacts of emissions-related legislation. For example,

[4]Note that the cost-push assumption in production is relaxed for energy inputs and traded inputs (see subsection 2.2 and subsection 3.2)

[5]The assumption of fixed coefficients is relaxed in the model [see section XX] as producers can substitute between energy carriers and between trade partners, including from and to domestic intermediate producers.

when simulating a domestic upstream energy price change relative to energy carriers' greenhouse gas ($CO_2eq$) content, such as through a carbon tax, the aggregate tax collected in the model need to correspond to the aggregate GHG emissions released in domestic production. In the context of economic modeling, this means that the price change applied to the monetary (or economic) flows of energy must be such that the product of the price change and the monetary flow will equal the product of the policy-implicated carbon price level ($USD/CO_2eq$), the volume of energy flow (in tonnes of oil equivalent, toe), and its emissions factor ($CO_2eq/toe$). If this is not the case, in this example, the carbon price that is (implicitly) modeled does not equal the carbon price of the policy scenario (cf. Hubacek et al., 2017 on this issue).

Hence, to map physical to monetary flows in the input-output data, a corresponding matrix of physical energy and energy-borne emissions flows is constructed, similar to Guevara and Domingos, 2017. The starting point is the basic input-output calculations described in subsection 2.1, with the exception that now, the focus is only on a subset of input sectors: Let $1 \dots e$ be fossil fuel sectors ($e \in n$). Hence, $Z_{emij}$ denotes the submatrix of direct fossil fuel inputs $e$ from country $i$ to all products $m$ ($e \in m$) produced in country $j$.

Ignoring cross-border flows for simplicity, the full direct inputs matrix $\mathbf{Z}$ can then be partitioned into energy and non-energy industries where the sub-matrices $\mathbf{Z}$ resemble a) the fossil fuel inputs to fossil fuel production (i.e., own-use, $Z_{ee}$), b) the (direct) energy inputs to non-energy industries $Z_{em}$, c) the non-energy inputs of fossil fuel industries $Z_{ne}$, and d) the non-energy inputs of non-energy industries $Z_{nm}$. Hence, the direct monetary flows of energy through the economy are depicted in submatrices $Z_{ee}$ and $Z_{em}$, which can be reconstructed into physical energy and emissions terms ($E$).

$$\mathbf{Z} = \begin{bmatrix} Z_{ee} & Z_{em} \\ Z_{ne} & Z_{nm} \end{bmatrix} \quad \longrightarrow \quad \mathbf{Z} = \begin{bmatrix} E_{ee} & E_{em} \\ Z_{ne} & Z_{nm} \end{bmatrix} \tag{8}$$

Total (global) energy flows can, thus, be described as:

$$Z_{e,e/m} + y_e = x_e \tag{9}$$

Similarly, all (monetary) direct energy flows to intermediate and final demand can be written as in Equation 9. The fuel production and consumption sectors in countries' energy balances data are mapped to the energy and non-energy sectors in the MRIO system to relate each monetary flow to the direct volume flow of energy for each fuel, or energy carrier. Fuel-sector-specific emissions factors are used to arrive at emissions flows.

### 2.2.1 Data

The MRIO-like matrices of fossil fuel and carbon dioxide ($CO_2$) emission flows are based on energy balances published by the International Energy Agency (IEA), the United Nations Department of Economic and Social Affairs (UNDESA), and Emissions Database for Global Atmospheric Research (EDGAR) data (Crippa et al., 2023; IEA, 2023; UNDESA, 2025), as well as fuel-, sector- and country-specific emission factors from the Greenhouse Gas and Air Pollution Interactions and Synergies (GAINS) model (IIASA, 2020). The full process, including codes and correspondence tables, is documented in detail as part of the Github repository.[6]

Based on sectoral and regional mappings (or concordance tables), the IEA country-level energy use data are aligned with GLORIA's structure and complemented with energy use data from the United Nations Statistical Division (UNSD). Energy balances are comprehensive datasets that track the supply, transformation, and final consumption of energy across all fuels and sectors within an economy, structured to ensure that total energy inputs equal total outputs by accounting for production, imports, exports, stock changes, conversion losses, and end-use consumption. For countries missing from the IEA and UNSD datasets, national energy use values are imputed using UNSD totals and reallocated across fuels and sectors in proportion to countries with similar income levels and energy structures. Regional aggregates are constructed by summing available country data and filling remaining gaps with these imputed estimates.

[6] https://github.com/cpmodel/energy-IEA; access upon request

A bilateral energy trade matrix (in physical units) is constructed using an iterative RAS procedure, ensuring consistency with known IEA row and column totals and using GLORIA's bilateral trade values (in monetary terms) as the initial seed. Physical trade flows are then disaggregated to the sector level using a sectoral fuel-use concordance, and further allocated across trading partners using GLORIA's bilateral trade shares by fuel and product. Special adjustments are made to account for fuels used in international transport (not included in domestic use totals) and for uses aggregated as "unspecified industry." Splitting energy flows from more aggregate IEA/UNSD consumption sectors to GLORIA industries assumes that (implicit) energy prices among those industries are equivalent (since they are split proportionately to monetary flows $z$). Between IEA consumption sectors, however, we allow for differentiated implicit prices.

Energy-borne emissions are calculated by applying fuel- and country-specific emission factors from the Greenhouse Gas and Air Pollution Interactions and Synergies (GAINS) model, and for fuels not covered in GAINS, standard emission factors for direct emissions from fuel combustion are used from the Intergovernmental Panel on Climate Change (IPCC) (IPCC, 2022). Non-combustion process emissions are estimated separately by distributing $CO_2$ emissions data from the EDGAR database (Crippa et al., 2023) across sectors using the output structure from GLORIA. The resulting harmonized database of energy use, bilateral trade in energy products, and sector-level emissions provides key inputs to MINDSET's analysis of climate mitigation and distributional impacts.

# 3 Model mechanisms

As a demand-led model, MINDSET estimates final demand components based on an endogenous structure. Labor demand leads to income generation, which, alongside price changes, drives final consumption of households and trade of intermediate and final demand. The resulting changes in sectoral output drive investment, or fixed capital formation, and finally, labor demand changes. As sectoral employment to output ratios change endogenously, the composition of value added is not constant. Similarly, the input-output coefficients for intermediate demand change endogenously with respect to energy inputs and traded partners. Endogenous changes are driven by a set of exogenously given elasticities and parameters (see Appendix Table 3).

The model interface is highly user-friendly and geared towards country-specific applications. The program is based on a modular framework with all parameter and scenario inputs given in amendable spreadsheets. The initial modules read and parse data and scenario inputs from various input spreadsheets. Once the shocks are formed, an iterative process is executed to solve the model, with different modules solved sequentially in each iteration loop[7]. As equations are not solved simultaneously, transmission channels are preserved in the model results. This means that the user can see which portion of a sectoral impact is driven by which channel or behavioral equation[8].

The sections below summarize the modules and main calculations that are included in each iteration. In this 1.0 version of the model, the calculations are limited or even fixed for some model variables. A dynamic version of the model in which all model variables carry values across time is completed and documented in the MINDSET 2.0 Model Documentation.

[7]The loop is a Gauss-Seidel-type iterative solution (Sauer, 2012, pp. 108-112) in which the system of equations is solved sequentially in a predetermined order. Once all equations have been covered, the process is repeated until differences across iterations are sufficiently small.

[8]For example, the user can see in the results to what extent manufacturing sector output was impacted negatively through higher energy prices and hence reduced domestic and international demand, but positively from increased investments in other sectors, and hence demand for manufacturing sector capital goods.

| Notation | Description | Notation | Description |
|---|---|---|---|
| $i$ | Importing regions | $j$ | Exporting regions |
| $m$ | Producing (origin) sectors | $n$ | Purchasing sectors |
| $k$ | Energy carrier ($k \in n$) | $g$ | Household consumption category ($g \in n$, COICOP group) |
| $t$ | Period (t0, t1) | $\Delta$ | Absolute change |
| $\delta$ | Relative (proportional) change | | |
| $Z$ | Matrix of intermediate inputs | $A$ | Technology (IO) matrix |
| $a$ | Technical coefficient (element of $A$) | $x$ | Total output vector |
| $E$ | Matrix of energy use (aggregate) | $e$ | Specific fuel use / energy carrier |
| $Y$ | Final demand (C + FCF + G) | $C$ | Household final demand / consumption |
| $G$ | Government consumption | FCF | Fixed capital formation (final demand) |
| $p$ | Producer price (sectoral) | | |
| $I$ | Investment by investing sector | $\alpha^{\text{INV}}$ | Investment converter |
| $Q$ | Sectoral output (production) | $L$ | Employment (sectoral) |
| $s$ | Bilateral import share | $P$ | CES composite price index (trade) |

Table 2: Commonly used notation

## 3.1 Final demand

Final demand is the sum of three main components, in line with national accounts (Lequiller and Blades, 2014): (1) household final demand ($C$), (2) fixed capital formation ($FCF$), and (3) government expenditures ($G$).[9] All three vectors are three-dimensional, differentiating the country of origin and the country of consumption for each sector (see Section 2.1). However, only household final demand and fixed capital formation (or investment) changes endogenously in MINDSET. While household consumption reacts endogenously to price and income changes, $FCF$ reacts to changes in sectoral output. Government spending changes only through exogenous scenarios, such as tax or public spending scenarios[10]. It is important to note that all three components can be forced to change endogenously in the case of a scenario which includes exogenous supply constraints (modeled through the Ghoshian upstream constraint, see Section 2.1). For example, if a drought destroys half of a country's wheat production, total final (and intermediate) demand for wheat supplied by this country is forced to drop by half. The remainder of this chapter describes endogenous changes to household consumption and investment ($FCF$).

### 3.1.1 Total final demand change

The sum of changes in household, investment, and government demand constitutes the total change in final demand $\Delta Y_{n,i,j}$ in country $i$ for products $n$ sourced from country $j$ (which may be the same as $i$):

$$\Delta Y_{n,i,j} = \Delta C_{n,i,j} + \Delta G_{n,i,j} + \Delta FCF_{n,i,j} \tag{10}$$

### 3.1.2 Household final demand

Household consumption ($C$) reacts primarily to changes in income and prices, reflecting observed household behaviors such as consumption inertia, habits, and limited substitutability between goods, as well as market

[9] Note that Non-Profit Institutions Serving Households (NPISH) is missing from the final demand components, given its small share in final demand and the non-market drivers inducing NPISH consumption. NPISH is, hence, assumed to be constant.

[10] Hence, government spending does not react to changes in GDP – and subsequent changes in government revenue. This can be interpreted in a way that governments do not (immediately) adjust spending.

imperfections. The level and composition of household final demand, or consumption, is driven by two main channels: (1) the household income effect, and (2) the household price effect. Beyond the general demand reaction to price changes, trade and energy substitution are addressed separately (see subsection 3.2 and subsection 2.2 on trade and energy substitution).

Household demand responses are informed by own-price, cross-price, and income elasticities of demand, taken from an international dataset of consumption elasticities (Meade et al., 2014; Muhammad et al., 2011). Elasticities are differentiated by 143 countries and by nine consumption groups, following COICOP (UNSD, 2018)[11]. For countries which are not separately listed in the elasticities estimates, aggregate estimates by country income group are applied, differentiated by low-, middle-, and high-income countries, from the same data source.

**Reactions to price changes** Consumption changes as a response to price changes are informed by own- and cross-price elasticities of demand. A sectoral concordance table maps the MRIO sectors, or products, $n$ to the nine groups, $g$. Hence, $n \in g$ indexes sectors within each consumption category $g$. MINDSET calculates changes in demand for products $n$ by applying a weighted average price change to each consumption group $g$.[12] Hence, first, the weighted average relative price change of consumption group $g$ in country $i$ needs to be calculated, $\delta p_{g,i}$.

$$w_{n(g),i,j} = \frac{C_{n,i,j}}{\sum_{n \in g,\, j} C_{n,i,j}} \quad (n \in g) \tag{11}$$

$$\delta p_{g,i} = \sum_{n \in g,j} \left( w_{n(g),i,j} \times \delta p_{n,i,j} \right) \tag{12}$$

where $\delta p_{n,i,j}$ is the relative price change of sector $n$ in country $i$, supplied from country $j$ ($i \in j$), and $w_{n(g),i,j}$ is the share of products (summing to 1) from sector $n$ produced in country $j$ in total consumption. $\sum_{n \in g,j} C_{n,i,j}$, of all products within consumption group $g$ in country $i$, irrespective of origin.

The changes in household consumption for all consumption categories $g$ in country $i$ sourced from country $j$ are then calculated in two steps, based on own-price and cross-price reactions.

The consumption effect due to own-price impacts ($\theta^{op}_{g,i}$) is calculated:

$$\theta^{op}_{g,i} = (1 + \delta p_{g,i})^{\varepsilon^{op}_{g,i}} \tag{13}$$

where $\varepsilon^{op}_{g,i}$ is the own-price elasticity of demand.

The absolute change in consumption of consumption category $g$ is based on cross-price reactions with each 'substitutable' consumption category $g$ ($g\prime \in g$). The overall change in consumption of $g$ is a simple average across all partial effects with substitutable items ($\theta^{\text{cp}}_{g,i}$):

$$\theta^{\text{cp}}_{g,i} = \frac{1}{|S(g)|} \sum_{g\prime \in S(g)} (1 + \delta p_{g\prime i})^{\varepsilon^{\text{cp}}_{g\prime \to g,i}} \tag{14}$$

where $|S(g)|$ is the number of categories $g\prime$ substitutable with $g$, the cross-price elasticities are $\varepsilon^{\text{cp}}_{g\prime \to g,i}$.

The overall price impact (in absolute terms) on consumption of $g$ in country $i$ $\Delta C^{price}_{g,i}$:

[11] The nine categories aggregate the twelve COICOP categories: Food, Beverages & Tobacco, Clothing & Footwear, Housing, House Furnishing, Medical health, Transport & Communication, Recreation, Education, and Other.

[12] This setup will be changed in a revision to MINDSET such that, instead, demand elasticities will be applied to each sector $n$ separately instead of to aggregate consumption groups.

$$\Delta C_{g,i}^{price} = C_{g,i}^{t_0} \times \theta_{gi}^{\text{op}} \times \theta_{g,i}^{\text{cp}} \tag{15}$$

where $C_{g,i}^{t_0}$ is the baseline consumption of goods basket $g$ in country $i$ ($C_{g,i} = \sum_j C_{g,i,j}$).

**Income effect** Household (disposable) income, $Y_i$, is defined as the sum of labor income from wages, $W_i$, (see Section 3.5) and social transfers minus income and a portion of payroll taxes given by the policy scenarios. Other sources of income, such as remittances or capital income, are not captured, implicitly keeping them constant, and households are expected to use additional income for consumption, not savings or investments.

Similar to price impacts, consumption changes as a response to changes in income are again calculated on the level of the consumption group $g$. The response to income changes is calculated in two steps, where the first step identifies the new income composition based on country- and consumption-group-specific income elasticities of demand, and the second step ensures that the aggregate consumption change aligns with the income change, imposing a budget constraint:

$$\Delta C_{g,i}^{inc} = C_{g,i}^{t_0} \times (1 + \delta y_i)^{\varepsilon_{g,i}^{inc}} \tag{16}$$

$$\tag{17}$$

Where $\varepsilon_{g,i}^{inc}$ is the income elasticity of demand for consumption category $g$ in country $i$, $C_{g,i}^{t0}$ is the baseline consumption of category $g$, and $\delta y_i$ is the relative change in household disposable income. The budget constraint is then imposed by scaling the total income-driven consumption change to the total income change, such that $\sum_g \Delta C_{g,i}^{inc} = \delta y_i \times Y_i^{t_0} = \Delta Y_i$.[13]

**Combined household final demand** The absolute group-level consumption impacts from price changes are then distributed across all products $n$ ($n \in g$) while keeping the product and origin shares fixed within each group:

$$\Delta C_{n,i,j} = w_{n(g),i,j} \times \left( \Delta C_{g,i,j}^{price} + \Delta C_{g,i,j}^{inc} \right) \quad (n \in g) \tag{18}$$

This setup assumes that within each consumption category $g$, the share of consumption of product $n$ remains unchanged (see Equation 12).

The demand elasticities governing responses to price changes are structurally unconstrained which has two implications regarding the consumption baskets as well as for the share of income implicitly allocated to savings. While consumption baskets evolve from the initial consumption structure, they lack any minimum subsistence share or aggregation condition (Deaton and Muellbauer, 1980), meaning that the model does not assume household budget exhaustion. This structure implies that the savings rate is endogenous, resulting from household consumption changes in response to prices over time.

To conclude, these modeling choices for household consumption provide a reduced-form structure for analyzing the distributional effects of fiscal policy or climate-related shocks in low- and middle-income countries. Consumption is specified as a function of changes in current income and prices - omitting any explicit wealth-holding or borrowing channel. This approach does not assume forward-looking behavior or intertemporal utility maximization; instead, it reflects bounded rationality, inefficient markets, and liquidity constraints (Banerjee and Duflo, 2007; Deaton, 1992).

[13] Note that there is no budget constraint for price-incuded consumptoin changes.

### 3.1.3 Fixed capital formation (investment)

Investment in MINDSET changes endogenously or exogenously through policy scenarios. Endogenous investment is purely a function of sectoral output changes. The absolute change in investment in sector $m$ in country $i$, $\Delta I_{m,i}$ , is:

$$\Delta I_{m,i} = \beta_{m,i}^{inv_output} \times \Delta Q_{m,i} \tag{19}$$

where $\Delta Q_{m,i}$ is the absolute change in output in sector $m$, country $i$, taking into account all endogenous and exogenous demand changes, and $\beta_{m,i}^{inv_output}$ is a country-sector-specific output investment factor specifying how much investment is induced from each unit of additional output.

Investments in sector m are then converted into final demand for capital goods produced by sector $n$ (and $m$) following an investment converter:

$$\Delta FCF_{n,i,j} = \Delta I_{m,i} \times \alpha_{m,n}^{INV} \times \frac{FCF_{n,i,j}^{t_0}}{\sum_n FCF_{n,i,j}} \tag{20}$$

where $\Delta\text{FCF}_{n,i,j}$ is the absolute change in the final demand for capital goods (and services) $n$ in country $i$ that originate from country $j$, $\Delta I_{i,m,t}$ is investment by sector $m$ in country $i$, multiplied by the relevant coefficient from the investment converter, $\alpha_{m,n}^{INV}$, and the proportion of the overall demand that is produced in country $j$. Changes in $FCF$ are proportionate to baseline flows, meaning that the ratio of imported to domestically produced capital goods is held constant.

Within the structure of input-output tables, $FCF$ is represented from the perspective of the producer of capital goods rather than the investor. Specifically, it appears as a final demand column, with the entry in each row indicating the value of capital goods produced by a given sector. The IO framework, therefore, does not provide information on the sector undertaking the investment but only on the origin of the capital goods.

In contrast, MINDSET conceptualizes investment from the standpoint of the investing sector. To operationalize this, the model employs an investment converter, $\alpha_{m,n}^{INV}$, (Cambridge Econometrics, 2022) that allocates sectoral investment expenditures to the capital-goods-producing sectors. For each investing industry, the converter defines the fixed proportion of capital goods sourced from any one sector, including itself.[14] In the specific case of exogenously defined electricity-sector investments, a separate converter informs how generation-technology-specific investments translate into FCF final demand. For instance, in contrast to investments in coal power plants, wind and solar photovoltaic investments will induce demand for electrical equipment.

### 3.1.4 Government

Government consumption, or spending, does not change endogenously. Hence, changes in GDP, and subsequent changes in revenues, are not expected to change government expenditures unless the policy scenario explicitly calls for it, e.g., that revenues collected through a tax scenario are 'recycled' towards specific sectoral government spending (or investment).

[14]The investment converter determines which sectors produce investment goods for other sectors based on Cambridge Econometrics, 2022. For example, the construction sector produces a large share of investment goods for the public sector, but the motor vehicles sector produces a large share of investment goods for transport services. The original data source is a set of bridging matrices published in the UK's national accounts. These matrices have been simplified to match the 120 sectors in MINDSET. In the conversion process, directly consumed capital goods remained unchanged, including machinery and construction-related sectors. Investment to demand conversion for the remaining shares is directed towards the sector investing, assuming that much of the remaining capital is produced within the sector (e.g., livestock, intangible capital, management of capital products).

## 3.2 Trade

The multi-regional input-output framework forms the basis of MINDSET's approach to international trade, capturing both inter-industry linkages and cross-border flows of intermediate and final demand. This structure provides a detailed picture of global supply chains, where goods and services pass through several countries before reaching final consumers.

Intermediate bilateral trade flows between industries in different countries respond to two main drivers: the initial structure of bilateral trade and relative sectoral price changes. Based on trade elasticities, import shares change, which results in changes in input coefficients (**A** matrix) while keeping technology constant. Hence, producing industries $n$ can switch suppliers, $m$, from one country to another (including domestic suppliers), but this will not alter their 'production recipes'. Trade in final demand products does not react to relative price changes between supplying countries, hence holding sectoral trade shares constant; only the overall composition of domestic versus imported production can change (see Section 3.1).

The reaction of intermediate bilateral trade flows to relative price changes is calculated in a multi-step process, using a CES (Constant Elasticity of Substitution) framework. First, baseline trade shares, as shares of total intermediate inputs, are calculated for each industry and country pair; then, reactions to price changes are aggregated across exporters to form a composite CES price index. This composite price is used to reweight each exporter's share in total intermediate inputs depending on its relative competitiveness. Finally, these updated input shares are multiplied by the origin-agnostic technical coefficient to produce new bilateral input–output coefficients that reflect substitution between trade partners:

The baseline bilateral share of sector $m$ used by producing industry $n$ in importer $i$ that comes from exporter $j$, $s_{m,n,i,j}$:

$$s_{m,n,i,j} = \frac{a_{m,n,i,j}}{\sum_j a_{m,n,i,j}} \tag{21}$$

where $\sum_j s_{m,n,i,j} = 1$ and $a_{m,n,i,j}$ is the input-output coefficient for inputs from sector $m$ in country $j$ to sector $n$ in country $i$.

The change in the composite CES price index for all inputs from sector $m$ to $n$ in country $i$ is then calculated as:

$$\delta P_{m,n,i} = \left[ \sum_j s_{m,n,i,j} \times (1 + \delta p_{m,i,j})^{1-\sigma_n} \right]^{\frac{1}{1-\sigma_n}} - 1 \tag{22}$$

where $\delta p_{m,i,j}$ is the relative price change in sector $m$ in country $j$ and $\sigma_n$ is the sector-specific trade elasticity.

The re-weighted exported shares as a reaction to all trade partners' price changes are a reaction to each import flow's price change weighted with the normalization factor:

$$\tilde{s}_{m,n,i,j} = s_{m,n,i,j} \times (1 + \delta p_{m,i,j})^{1-\sigma_n} \times (1 + \delta P_{m,n,i})^{\sigma_n - 1} \tag{23}$$

where $\delta p_{m,i,j}$ is the sectoral price change in the supplying country $j$.

Then, lastly, the updated input-output coefficients for all intermediate flows are calculated as:

$$a^{t1}_{m,n,i,j} = a_{m,n,i} \times \tilde{s}_{m,n,i,j} \tag{24}$$

$$a_{m,n,i} = \sum_j a_{m,n,i,j} \tag{25}$$

Using the origin-agnostic technical coefficient of $m$ into $n$ for importer $i$, $a_{m,n,i}$ to ensure producers are switching between suppliers, but not changing their input structures.

### 3.3 Energy efficiency and energy substitution

MINDSET captures energy use and energy-borne GHG emissions in physical and monetary terms (see Section 2.2 for an overview of how the data are constructed), allowing for modeling energy- or emissions-based policy scenarios, as well as their economic and environmental impacts across the economy. Policy scenarios can be differentiated by energy carrier and using sector. This means that, e.g., changes in excise taxes on natural gas can be differentiated between the electricity and manufacturing sectors. In response to the exogenous price changes across energy carriers, industries will undertake energy efficiency measures or switch between fuels. MINDSET, hence, allows for endogenous changes in input-output coefficients, or production recipes, with respect to direct energy inputs, including fossil fuels and electricity.

It is important to note that energy use levels in MINDSET scenarios can also be informed by exogenous inputs from an energy-sector model, such as the TIMES model, or an electricity-sector model, such as the World Bank's Electricity Planning Model (EPM) or FTT:Power (Mercure, 2012).

Exogenous own- and cross-price elasticities (Stern, 2012) for direct energy inputs govern the price-induced energy efficiency and energy substitution adjustments. The input structure of industries $n$ in country $i$ with respect to energy inputs $k$ ($k \in m$) subject to a price change in country $j$ (again, $i \in j$). The price effect on technical coefficients induced by the own price elasticities $theta^{op}_{k,n,i,j}$ is:

$$\theta^{op}_{k,n,i,j} = (1 + \delta p_{k,n,i,j})^{\varepsilon^{op}_{k,i}} \tag{26}$$

The price changes of energy inputs $k$ to energy and non-energy sectors $n$, $\delta p_{k,n,i,j}$, and the own-price elasticities $\varepsilon^{op}_{k,i}$, $\varepsilon^{op}_{k,i} < 0$, govern the energy efficiency response[15].

Energy substitution, or fuel switching responses, are calculated based on cross-price elasticities for all substitutable fuels $k\prime$ ($k\prime \in k$), using cross-price elasticities $\varepsilon^{\text{cp}}_{k\prime \to k,i}$ between substitutable fuels $k\prime$ in producing countries $n$. The cross-price effect on technical coefficients between fuel inputs $k$ and producing sectors $n$ across all partial effects of substitutable fuels is $\theta^{\text{cp}}_{k,n,i,j}$:

$$\theta^{\text{cp}}_{k,n,i,j} = \frac{1}{|S(k\prime)|} \sum_{k\prime \in S(k)} (1 + \delta p_{k\prime,n,i,j})^{\varepsilon^{\text{cp}}_{k\prime,k,i}} \tag{27}$$

Where $|S(k\prime)|$ is the number of energy carriers $k\prime$ substitutable with $k$, such that the combined effect $\theta^{\text{cp}}_{k,n,i,j}$ is the simple average over substitutable fuels $k\prime$ as a reaction to their price change $\delta p_{k\prime,n,i,j}$.

The combined effect of both own- and cross-price elasticities is then:

$$\delta a_{k,n,i,j} = \theta^{op}_{k,n,i,j} \times \theta^{cp}_{k,n,i,j} \tag{28}$$

It is important to note that, in practical terms, there is no actual price increase of the fuel itself, but the cost is imposed on the use of the fuel. Hence, the costs are added to the input costs of downstream production based on the new t1 direct energy input flows $Z^{t0}_{k,n,i,j} \times \delta a_{k,n,i,j}$. If energy price increases are due to a policy change, such as in a carbon pricing, energy tax, or subsidy scenario, fiscal revenues are calculated based on the new t1 direct energy input flow matrix $\mathbf{Z}^{t1}$ and the fuel-sector-specific price increases $\delta p_{k,n,i,j}$, hence, after energy efficiency and substitution.

[15] All default parameter inputs or elasticities in the model are exogenously given through Microsoft Excel files. Hence, they can easily be change in their magnitude and also in their dimensionality, with some additional code adjustments.

## 3.4 Prices

Price mechanisms in MINDSET are limited, and price changes mainly depend on scenario inputs. Prices are determined on the supply side, with demand agents primarily acting as price takers, adjusting only their consumption patterns in response to price changes. Consistent with the MRIO framework (see Section 2.1), price effects are differentiated into direct effects and indirect effects. Direct price effects arise from cost changes *within* a given sector, such as due to changes in value-added costs, labor or capital, or direct taxation. Indirect price effects capture the propagation of these direct price changes along supply chains, based on the intersectoral and trade linkages embedded in the MRIO structure. In MINDSET, only the indirect price changes propagate endogenously. Direct price changes in the model occur due to price-related policy scenarios such as subsidy or carbon pricing reforms, changing the cost of the product itself (in this case, energy), or payroll tax changes, which change the cost of value-added inputs, in this case, labor.

Direct price changes can also be based on user assumptions; for example, if private sector adaptation investment costs, such as for air conditioning, are expected to be passed on to demand agents, the expected direct price increases by the investing sector need to be calculated exogenously and included as part of the scenario. As there are no profit margins in MINDSET, direct tax or price impacts on production are expected to be fully passed on, e.g., the price of coal will rise proportionally to the modeled coal tax increase.

Indirect price changes then occur based on the Leontief price model (see Section 2.1). With the exception of direct inputs of energy and imports, price changes of intermediate inputs are fully passed through along supply chains. For energy and traded inputs, price increases can be mitigated through energy efficiency, fuel switching, and switching between trade partners or to/from domestic suppliers (see Section 2.2 and Section 3.2).

## 3.5 Employment and labor income

Sectoral employment is modeled as responding to changes in sectoral output, governed by sector- and country-specific output-employment coefficients. Hence, the relative change in employment in sector $n$ in country $i$ is $\delta L_{i,n}$:

$$\delta L_{i,n} = \beta_{i,n}^{labor} \times \delta Q_{i,n} \tag{29}$$

While this simplification cannot capture endogenous labor substitution or technological change, it provides a transparent and empirically tractable basis for estimating employment impacts. The default dataset can easily be updated with revised estimates for countries of interest. In the longer run, this relationship can be extended by incorporating exogenous trends in labor productivity.

Wages in MINDSET are fixed. Hence, the economy-wide compensation of employees changes proportionally with employment:

$$\Delta W_i \; = \; \sum_n W_{i,n}^{t_0} \times \delta L_{i,n} \tag{30}$$

Hence, in the default specification, increases in labor value added are absorbed entirely through changes in employment, with average wages fixed at their baseline level. In some applications, however, this assumption has been relaxed to allow for value-added gains to be shared between expanding employment and higher wages for existing workers.[16]

[16] Examples of this are the Lao PDR and Mongolia CCDRs where MINDSET results were used to run the World Bank's Microsim or 'GIDD'; see subsection 6.1 for CCDR sources.

# 4 Distributional effects for workers and consumers

MINDSET's estimates of sectoral changes in consumer prices and labor demand serve as valuable inputs for macro-micro, or top-down, microsimulations to understand policies' employment and consumption equality effects. The model is particularly well-placed to inform distributional outcomes as it captures the short-term frictions and related transitional costs associated with policy-induced sectoral reallocations. As a standalone tool, MINDSET disaggregates results across a wide range of economic sectors. However, this conceals the underlying equity effects across socioeconomic strata as well as the resulting changes in poverty levels. Hence, the model results serve as input for top-down macro-micro simulations such as the World Bank's Microsim model for poverty outcomes (Bourguignon, Bussolo, and Pereira da Silva, 2008)[17] or the World Bank's Commitment to Equity (CEQ) analysis for consumer price and social assistance distributional impacts. In addition, MINDSET results can be combined with the model's own microsimulation modules, the Employment and the Consumption Modules.[18] The shorter-term distributional effects of employment and consumption effects are measured through augmenting MINDSET's results with countries' most recent microdata. Countries' labor force and household budget survey data are harmonized following a common MINDSET codebook [19], capturing all relevant socioeconomic indicators such as disposable (or market) income, consumption patterns, occupation and sector of work, gender, subnational region, rural/urban.[20] The sectors of work – or for consumption impacts, the product categories - serve as the linking variable between MINDSET results and the microdata. In their current version, there is no feedback from the microsimulations back to MINDSET. This chapter gives an overview of both modules.

## 4.1 Employment Module

The Employment Module provides a simplified version of macro-micro simulation models, focusing on the short- to medium-term distributional effects of policy-induced labor demand changes and how they impact different socioeconomic groups. As in more sophisticated macro-micro models, the main linking variable between the macroeconomic model and the microsimulation is the vector of sectoral employment changes. The microsimulation is based on countries' labor force surveys (LFS), such as from the World Bank's Global Labor Database (GLD), or similar individual-level data representative of countries' populations. These are used to identify the average characteristics of workers (employed or self-employed) in each sector. Collected characteristics include the level of income[21] and respective income decile, occupation, age, gender, and region or province of work.[22]

For each group $d$ of average workers in sector $n$ in country $j$, the estimated employment impact $\Delta l$, then, follows from the MINDSET results:

$$\Delta l^{d}_{n,j} = \delta N_{j,n} \times lf^{d}_{n,j} \tag{31}$$

where $\delta N$ is the relative change in sectoral employment in sector $n$ compared to the baseline, according to MINDSET results, $lf^{d}_{j,n}$ is the number of workers of group $d$ working in sector $n$, and group $d$ can be characterized by any combination of the socioeconomic characteristics of interest (listed above).

[17] For example, the poverty impacts of climate change in Laos were estimated by combining the two macro and micro models (with MINDSET) (World Bank (forthcoming))

[18] Codes, concordance tables and documentation available on Github (https://github.com/cpmodel/mrio-microdata; access upon request).

[19] Available on Github (https://github.com/cpmodel/mrio-microdata; access upon request)

[20] To maximize synergies with other tools, MINDSET's Consumption Module follows the same harmonization of microdata as the IMF-WB Climate Policy Assessment Tool (CPAT).

[21] Income definition (disposable, market, or labor) depends on availability in a country's microdata. Preference is given to disposable income.

[22] The MINDSET Codebook guides the harmonization of each country's survey in accordance with the Employment Module's coding program (both available on GitHub). Harmonizing an LFS requires no more than one day of work. The resulting multi-country harmonized dataset supports ad-hoc requests from a wide range of countries.

Importantly, this method is static and is based on certain assumptions which make it particularly suitable for short- to medium-term distributional analyses. It is static in the sense that it takes as inputs only one year of MINDSET results, not a time series of sectoral effects. Being a static approach, it rests on two important assumptions: first, the labor force composition remains unchanged up until the year of interest. This means the composition of workers in each subsector remains the same as in the survey year, in terms of their relative positioning in the income distribution, gender, age, and region of work etc.. As a consequence, the Employment Module is appropriate for short- to medium-term distributional analyses. Second, the profile of new jobs and of the workers taking them is assumed to follow the same characteristics as the existing jobs and workers in each sector. As a consequence, the analysis is less suitable for policy analyses which involve large within-sector transitions, e.g., a large-scale shift from coal-fired to renewable electricity, which would likely require workers of different skills and education. Arguably, such shift would anyway occur over the longer term and their distributional analysis would require a level of occupational detail in the survey data which rarely exists. These two assumptions leave the Employment Module highly suitable for analyzing the whole-of-economy distributional effects of any standard development policy package. [23]

The data requirements are moderate, enabling a low-cost, yet highly granular analysis suitable for application in almost all low-income countries. The Module requires only one year's labor force survey. Yet, the analysis provides a high level of socioeconomic granularity. Survey data are being harmonized to the international standards (specifically, to the International Standard Industrial Classification of All Economic Activities (ISIC) Revision 4 and the International Standard Classification of Occupations (ISCO) 2008 revision). If surveys permit, the linking variable, sector of work, is differentiated by 88 sectors (ISIC Rev. 4 2-digit).

The highly granular analyses provide important insights for policy advice on labor market support measures as well as social policies by identifying losers (or losing groups) and winners, as well as potential skill mismatches. Because MINDSET takes into account the frictions which occur during sectoral and job reallocation instead of solely eliciting the new equilibrium outcome, model results provide information on both the winners and losers *during* the process of reallocation:

On the one hand, MINDSET's Employment Module helps to identify socioeconomic groups that might need particular support during a transition or due to climate change impacts. Job losses from policy or changing weather and hydrological patterns are often concentrated among specific socioeconomic groups. While some of these impacts might be foreseeable without additional modeling (such as the impact of climate change on low-income subsistence farmers), other similarly concentrated job losses might not be as apparent; for example, the loss of relatively well-paying jobs of low-skilled women in hospitality industries due to a reduction in tourism driven by increasing temperatures.

On the other hand, understanding the occupational and sectoral composition of new jobs can help identify which labor market support policies might be needed to seize the economic opportunities presented by a given development policy package, and to avoid economic slowdown due to skill mismatches (such as is the case in Europe McGrath (2021) and Di Pietro (2024). Based on detailed information on a country's labor force composition, the Employment Module elicits which sectors and occupations could see the largest labor supply bottlenecks and skill shortages for achieving a given policy objective. Hence, it can provide the basis for informing targeted labor market support measures. For instance, resilient development strategies often require sizable investment in new and better infrastructure which can lead to spikes in demand for medium-skilled craft and trade-related workers in construction (World Bank, 2022a).

## 4.2 Consumption Module

The Consumption Module estimates the distributional effects of both tax incidence and revenue recycling into direct household income support or infrastructure investments in a narrow monetary sense, proxied by relative changes in consumption. The method follows established approaches to consumption-focused microsimulation (Dorband et al., 2019; Vogt-Schilb et al., 2019; Hubacek et al., 2017). Country-specific household budget surveys (HBS) are combined in a consistent manner with MINDSET's macroeconomic results

[23] Since the distributional routines in MINDSET are simplified, the sectoral and labor market detail is intended to facilitate linkages with more sophisticated tools such as the World Bank's Microsim model (Bourguignon, Bussolo, and Pereira da Silva, 2008).

on price changes and, where applicable, tax revenues. The main linking variable between the macroeconomic model and the microsimulation is the vector of sector-specific price changes, and the aggregate amount of government revenues directed to direct household income support. Countries' HBS are harmonized according to the MINDSET Codebook to identify average characteristics of households across socio-economic groups. To maximize synergies with other World Bank tools, MINDSET's Consumption Module follows the same harmonization of data as the IMF-WB Climate Policy Assessment Tool (CPAT), focusing on disposable income deciles and consumption patterns of goods and services, and of different fuels and electricity.

For the **incidence from consumer price changes**, the HBS consumption aggregates need to be scaled to the MINDSET's household final demand vector, to ensure aggregation consistency between the macro and the micro results (cf. Dorband et al., 2019; Hubacek et al., 2017. A concordance table is used to match model-level products $n$ to HBS-level consumption categories $g$. For each consumption category $g$ from the household survey, the procedure allocates model-level products $n$ ($n \in g$) into consumption category totals for each income decile $d$. Jointly with data on countries' total population in the year of interest (World Bank, 2024), this returns per-capita baseline consumption by consumption category $g$ and decile $d$ such that:

$$C^d_{g,i} = \sum_{n \in g} C^d_{n,i} \tag{32}$$

Similarly, relative price changes $\delta p_{g,i}$ are calculated as the weighted average price change of products $n$ in $g$, using the household final demand vector entries as weights.

$$\delta p_{g,i} = \sum_{n \in g} w_{n(g),i} \times \delta p_{n,i} \tag{33}$$

Where $w_{n(g),i}$ is household final demand vector entries as weights of n in g. The immediate overall consumption incidence, *not accounting for households' behavioral reactions* to price changes, can then be calculated for each decile $d$:

$$\delta C^d_i = \frac{\sum_g \delta p_{g,i} \times C^d_{g,i}}{\sum_g C^d_{g,i}} \tag{34}$$

The consumption incidence *with behavioral reactions* is, in the current version, calculated as:

$$\delta C^d_i = \frac{1}{C^d_i} \sum_g \delta p_{g,i} \times C^d_{g,i} \times (1 + \delta p_{g,i})^{\varepsilon^d_{g,i}} \tag{35}$$

Where $\varepsilon^d_{g,i}$ is the decile- and category-specific own-price elasticity of demand. The price elasticities of demand are taken from the literature (Meade et al., 2014; Muhammad et al., 2011), but can be adjusted to reflect MINDSET's own elasticities.

The incidence analysis of the **direct household income support** is based on country-level population data (World Bank, 2024). Let total population be $POP_i$, total revenue earmarked for direct household support be $R_i$, and let $d^* \in \{1, \ldots, 10\}$ be the highest targeted decile (so deciles $d^* \leq d$ are eligible). Thus, the targeted population is:

$$POP_{d^*,i} = \left(\frac{d^*}{10}\right) POP_i \tag{36}$$

The uniform per-capita transfer is then given by:

$$\tau_i = \frac{R_i}{POP_{d^*,i}} \tag{37}$$

Lastly, the Consumption Module allows for mapping **gains from investments in infrastructure provision** to the household level, following the approach developed in Dorband et al. (2022), also deployed in the IMF-WB CPAT. Government investments which are earmarked for infrastructure provision - water, sanitation, electricity, information and communication services, and public transport – can be simulated across consumption deciles as well. For each infrastructure category, the funds can be allocated across households without access in the baseline. Decile-specific "no-access" populations are obtained by applying access rates to population counts, where the latter are taken from countries' HBS. The allocation of potential, albeit non-monetary, 'benefits' is approximated through assumed 'transfers': For each infrastructure category, per-capita transfers are defined as the total allocation divided by the total baseline "no-access" population; the decile-level amount is then based on each decile's share of households without access. Summation over categories yields the total public-investment transfer for each decile. This approach to estimating the benefits of infrastructure access transfers abstracts in a number of ways, which are discussed in detail in Dorband et al. (2022).

Among others, there are two important limitations which apply to both income- and infrastructure-based revenue recycling options: The simplified method implicitly assumes perfect targeting, i.e., that all eligible households receive support, no leakage, i.e., that funds get fully spent for the identified purpose, and no administrative costs, i.e., that the full amount of revenues is available for redistribution. Furthermore, beyond the level of household income, the method does not consider the targeting or eligibility criteria of existing social assistance programs and other direct household support. For designing the specifics of social security or assistance reforms, complementary microsimulation models, which can account for these features, are available, such as the World Bank's Commitment to Equity (CEQ) analysis tools.

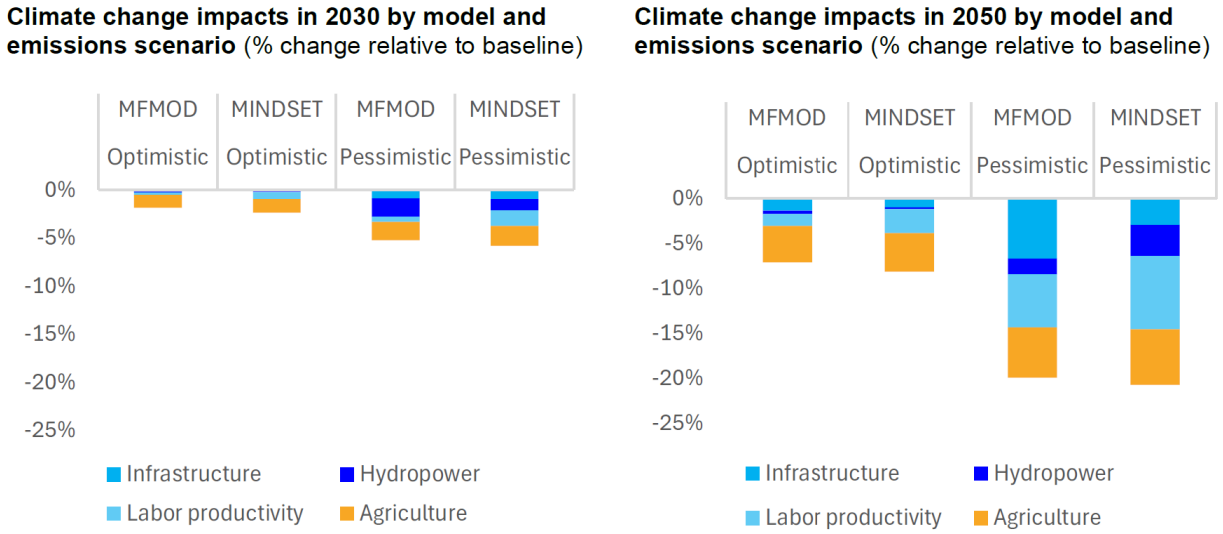
# 5 Discussion

This section provides a brief, non-exhaustive discussion of MINDSET 1.0 (MINDSET hereafter).

## 5.1 Model comparisons

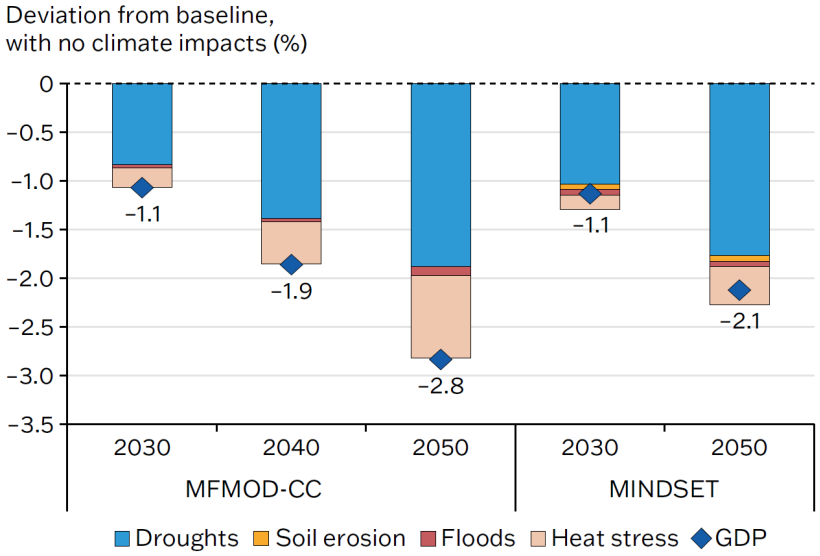
The goal of the model is to provide a simple, highly cost-effective (or 'rapid'), and tractable framework for understanding the granular distributional impacts across industries and people of detailed, actionable policy scenarios in specific country contexts. However, this approach also needs to be validated in terms of macroeconomic results. If the model's macroeconomic results are unreliable, the estimated distributional impacts might also not be in line with what might be expected.

Validating macroeconomic models against complex policy scenarios is very difficult, as the counterfactual cannot be established, but inter-model side-by-side comparison provides an alternative approach. While deploying alternative models for policy scenario analysis is best practice, not least to show a range of potential outcomes and how these might depend on the models' assumptions of economic behavior (World Bank, 2022b), it needs to be emphasized that differences between model results are to be expected - the whole point of deploying models side-by-side (Hartvig et al., 2025). Yet, in contexts where model inputs were highly streamlined and fully comparable between two models, comparison of MINDSET macroeconomic results with MFMOD's macroeconomic results has shown robust comparability between the two. This holds across country contexts and modeled scenarios (see Figure 1)




Source: World Bank analysis using MFMod and MINDSET models.

**Figure 4.1. Climate change impacts on GDP in the absence of adaptation measures**




*Source:* World Bank staff calculations, using MFMOD-CC and MINDSET.

*Note:* The aggregate effects observed are smaller than the sum of the individual climate impacts on GDP, when considered separately, because heat stress–related productivity losses in agriculture are smaller when the sector has already shrunk due to drought.

Figure 1: Comparison of model results between MINDSET and MFMOD (Burns et al., 2019) when using exact same assumptions; examples from Lao PDR CCDR, Figure 2.5, and Moldova CCDR, Figure 4.1 (see Appendix 6.1)

## 5.2 Limitations

### 5.2.1 Adjusting default data and parameters for country applications

Deploying MINDSET in a country engagement requires checking and possibly improving data and parameter assumptions. In its global default version, MINDSET relies on a global MRIO database as well as parameter estimates taken from the literature. User-friendly Microsoft Excel spreadsheets facilitate the updating of input parameters.

The use of multi-regional input-output data for single- and multi-country policy analysis comes with certain caveats and limitations, which may necessitate prior data improvements in certain countries. The main constraint common to all MRIO systems is that they undergo matrix balancing to ensure that updated row and column totals align with known aggregate values while preserving the structural relationships of the original input–output data. Because data quality is usually higher in larger countries, small countries can be most affected by the balancing. Other limitations are specific to the MRIO system in use, e.g., the data can be more or less closely aligned with countries' input-output tables (including the Z matrix of intermediate input flows). For these reasons, applying the model to a specific country context often requires checking and improving the country's data within the GLORIA system.

Similarly, model users are encouraged to update model parameters, which might be particularly important for the policy or shock scenario of interest. For example, in scenarios with energy price changes, parameters which could be important to update the own- and cross-price elasticities of intermediate energy inputs as well as the elasticities governing household demand. Country-specific parameters can either be estimated based on country data or taken from the literature.

### 5.2.2 Exogenous price adjustment, constraints, and the burden on model users

MINDSET strictly assumes economic slack in the baseline, which puts an additional burden on the model user when considering contexts or scenarios where capacity constraints might be binding. With direct price changes being exogenous in MINDSET, the underlying assumption is that additional economic resources, such as labor and capital stock (not necessarily financing), can be mobilized if demand for higher production is there. The assumptions of involuntary unemployment, or underemployment and resulting wage rigidities,[24] as well as underutilized production capacities,[25] are particularly suitable for lower-income country contexts characterized by large informal economies.

There are contexts and scenarios, however, where capacity constraints are binding and model users are tasked to account for this through exogenous scenario inputs. Examples are acute climate shocks, such as floods, preventing people from reaching their work and factories from being used or even destroyed. In these scenarios, the model user is tasked with implementing the expected rationing of production output. For example, depending on the share of a sector's capital affected and the duration of the flood as well as the time it takes to re-build the destroyed capital, the used needs to calculate off-model the expected shortfall of the sector's production as well as the costs for reconstruction. Losses of production will then enter the model as hard upstream constraints on production, modeled through the Ghoshian quantity model (see Section 2.1), which affects downstream sectors through supply shortages depending on the criticality of the specific input in their production process (Pichler et al., 2022). Rebuilding costs, if borne by the private sector and expected to be passed on in higher product costs, need to be included in the scenario in terms of exogenous direct price increases in the affected sector.

Another aspect that requires substantial exogenous assumptions is using the model for long-term analyses. As is the case in other models, assumptions need to be made regarding the aggregate baseline growth path

[24] While labor markets in low-income countries remain poorly understood as they are often driven by local specificities of institutional rigidities and norms, there is increasing evidence suggesting that informal labor markets in developing countries feature substantial involuntary unemployment with much of "self-employment" masking rationing and underemployment (see, e.g., Breza and Kaur, 2024 on rural India

[25] Firms in the formal economy in Europe rarely go above a reported 90% capacity utilization, with an estimated 80-85% long-term capacity utilization rate in the Europe (Pollitt et al., 2017; European Commission, 2024). Capacity utilization rates are typically much lower in lower-income economies (Walker et al., 2024; Egger et al., 2022).

of the economy as well as expected baseline structural changes, requiring the re-weighting of sectoral splits (mainly between agriculture, services, and industry).

# 6 Appendix

## 6.1 List of WBG publications deploying MINDSET

Published or forthcoming only (as of September 2025):


World Bank Group. 2022. Philippines Country Climate and Development Report. CCDR Series;. World Bank, Washington, DC.

World Bank Group. 2022. Peru Country Climate and Development Report. CCDR Series;. World Bank, Washington, DC.

World Bank Group. 2022. Bangladesh Country Climate and Development Report. CCDR Series;. World Bank Group, Washington, DC.

World Bank Group. 2022. Climate and Development: An Agenda for Action - Emerging Insights from World Bank Group 2021-22 Country Climate and Development Reports [Figure 13]. World Bank, Washington, DC.

World Bank. 2022. Green Fiscal Reforms: Part Two of Strengthening Inclusion and Facilitating the Green Transition. EU Regular Economic Report; No.7. World Bank, Washington, DC.

World Bank Group and Bank Negara Malaysia (BNM). 2022. Managing Flood Risks: Leveraging Finance for Business Resilience in Malaysia. World Bank, Washington, DC.

World Bank Group. 2023. Cambodia Country Climate and Development Report (English). CCDR Series. World Bank Group, Washington, DC.

Hallegatte, Stéphane; Godinho, Catrina; Rentschler, Jun; Avner, Paolo; Dorband, Ira Irina; Knudsen, Camilla; Lemke, Jana; Mealy, Penny. 2023. Within Reach: Navigating the Political Economy of Decarbonization. Climate Change and Development Series. Washington, DC: World Bank.

Lehr,Ulrike Gabriel Elisabeth Augusta; Pollitt,Hector Ben.2024. Heading Towards 1.5$^{o}$C - Impacts on Labor Demand in Selected Countries. Washington, D.C. : World Bank Group.

World Bank Group. 2024. Bosnia and Herzegovina Country Economic Memorandum, January 2024. Washington, DC: World Bank.

World Bank. 2024. The impact of the implementation of EU's Carbon Border Adjustment Mechanism on North Macedonia. Washington, DC: World Bank.

World Bank Group. 2024. Mongolia Country Climate and Development Report. Washington, DC: World Bank.

World Bank Group. (2025). Vietnam 2045 Growing Greener. Washington, DC: World Bank.

World Bank Group. (forthcoming) Laos Country Climate and Development Report. CCDR Series. Washington, DC: World Bank.

World Bank Group. (forthcoming) Malaysia Country Climate and Development Report. CCDR Series. Washington, DC: World Bank.

World Bank Group. (forthcoming) Fiji Country Climate and Development Report. CCDR Series. Washington, DC: World Bank.

World Bank Group. (forthcoming) Kyrgyzstan Country Climate and Development Report. CCDR Series. Washington, DC: World Bank.

World Bank Group. (forthcoming) Namibia Country Climate and Development Report. CCDR Series. Washington, DC: World Bank.

World Bank Group. (forthcoming) Botswana Country Climate and Development Report. CCDR Series. Washington, DC: World Bank.

International Finance Corporation (IFC) (forthcoming) A Gender Inclusive Transition: Opportunities and Risks for Women's Employment through Green Investment. Washington, DC: IFC.

World Bank Group. (forthcoming) Carbon Pricing to Support Low Carbon Transition in Indonesia Washington, DC: World Bank.

## 6.2 Data and parameter sources

Table 3: Data and Parametrization (global defaults, subject to revision)

| **Component** | **Dimensions** | **Source** |
|---|---|---|
| Multi-Regional Input-Output data | 120 sectors & 164 countries/regions | GLORIA MRIO database (Lenzen et al., 2017; Lenzen et al., 2021) |
| Energy substitution/cross-price elasticities | sector, country & fuel type | Stern, 2012 |
| Household price & income elasticities of demand | income group & consumption category | Meade et al., 2014 |
| Trade elasticities | sector & country | Sager, 2020 |
| Output-employment elasticities | sector & country | Own calculations or Cambridge Econometrics, 2022 |
| Output-investment factors | sector & country | Cambridge Econometrics, 2022 |
| Investment-FCF converter | sector pairs & country | Cambridge Econometrics, 2022 |

## 6.3 Software

MINDSET is a combination of (1) pre-processed datasets in Python's pickle format, CSV, and Microsoft Excel files, (2) a simulation program written in the Python programming language that is the core of the model, and (3) scenario files set up in Microsoft Excel. To ensure complete reproducibility, the full model, including data processing and model codes, parameter and data inputs, as well as concordance tables, is documented in detail in its Github repository[26]. To facilitate accessibility further, all model dimensions can be 'collapsed', turning the 163-country version into a user-defined number of countries and aggregate regions of interest. Hence, MINDSET can be run even on desktop computers with limited computational capacity; however, running the full version of the model (163 regions with bilateral linkages across 120 sectors) requires significantly more computing capacity.

[26] `https://github.com/cpmodel/MINDSET_module`; access upon request.

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